QUANTEDGE 1.0 FINAL CHALLENGE — COMPETITION BRIEFING

Quantitative Portfolio Construction, Extreme Value Theory & Multi-Layer Risk Defense

Institutional Investment Memorandum for The Serendib University Endowment Fund (USD 10 Million Mandate)

Institutional Client:	The Serendib University Endowment Fund ($10,000,000 USD Mandate)
Client Profile & Goal:	Conservative institutional investor seeking capital preservation, income, and real growth (CPI+4%).
Core Philosophy:	*'Avoiding large losses is more important than maximizing gains. A good decision has tested, defensible downside.'*
Investable Universe:	7 ETFs: SPY (US Equities), EFA (Dev ex-US), EEM (Emerging), AGG (US Bonds), GLD (Gold), XLE (Energy), XLK (Tech)
Core Methodology:	Markowitz MPT + 3 VaR Models + Expected Shortfall (CVaR) + Extreme Value Theory (EVT/POT) + Backtesting
Recommendation:	**ACCEPT MANDATE WITH EVT TAIL BUFFERS** (40% AGG, 20% SPY, 15% GLD, 10% EFA, 5% XLE, 5% XLK, 5% EEM)

EXECUTIVE SUMMARY & MANDATE DEFENSE

This memorandum presents the quantitative architecture developed by our research division to manage the **USD 10 Million endowment mandate**. Guided by the university's priority of capital preservation, we deploy an end-to-end framework integrating Modern Portfolio Theory, three distinct Value-at-Risk (VaR) engines, coherent Expected Shortfall (CVaR), and Peaks-Over-Threshold Extreme Value Theory (EVT).

Key Conclusions for the Investment Committee:

1. **Fulfills Real Spending Hurdle:** The recommended Conservative Endowment allocation delivers an annualized expected return of **+7.89%** with a Sharpe ratio of **0.61** and low volatility (**10.40%**), achieving the university's target spending rate of CPI + 4.0% (~6.5-7.5%).
2. **Superior Crisis Drawdown Mitigation:** The portfolio restricts maximum peak-to-trough drawdown to **-28.52%** over 20 years (including the 2008 GFC and 2020 COVID crash), significantly outperforming the standard 60/40 benchmark (-34.70%) and the S&P; 500 (-55.19%).
3. **Uncovering the Fatal Gaussian Tail Gap:** At 99% confidence, conventional normal distribution models underestimate tail losses by **+60.5%**. Our Generalized Pareto EVT model captures heavy tail dynamics (ξ=0.2440), providing the committee with an empirical worst-case loss ceiling of 2.75% ($275,160).
4. **Flawless Regulatory Validation:** In out-of-sample backtesting across 2020–2026 (N=1,697 days), the EVT model logged 16 violations against 17 expected (Kupiec p=0.8149), qualifying the portfolio for the supervisory **Basel GREEN Zone**.

1. Institutional Client Profile & Investment Philosophy

The Serendib University Endowment Fund manages a perpetual capital pool of USD 10 million. The fund's fiduciary purpose is to generate sustainable real income to fund student scholarships, endowed professorships, research initiatives, and strategic capital expenditure. Unlike commercial hedge funds or retail wealth managers, university endowments are subject to strict intergenerational equity constraints: the purchasing power of the endowment must be preserved indefinitely across decades of inflationary regimes and market dislocations.

The Investment Committee's core philosophy dictates that *'Avoiding large losses is more important than maximizing gains'*. A 50% drawdown requires a 100% gain merely to break even, inflicting irreparable damage on multi-year scholarship commitments. Accordingly, our objective function is not raw return maximization, but the maximization of downside-adjusted compounding and tail resilience.

2. Investable Asset Universe & Strategic Role Justifications

In strict compliance with Section 5 of the Competition Brief, the portfolio is constructed from the prescribed seven liquid ETF building blocks. Table 1 outlines the macroeconomic justification and portfolio role of each asset:

Ticker	Asset Description	Asset Class	Role in Endowment Mandate	Weight (%)	Allocation ($USD)
AGG	iShares Core US Aggregate Bond	US IG Fixed Income	Core capital preservation anchor; reliable coupon cash flow.	**40.0%**	$4,000,000
SPY	SPDR S&P; 500 ETF Trust	US Core Equity	Primary engine for long-term real purchasing power compounding.	**20.0%**	$2,000,000
GLD	SPDR Gold Shares	Commodity / Real Asset	Non-correlated safe haven, currency hedge, and tail risk buffer.	**15.0%**	$1,500,000
EFA	iShares MSCI EAFE ETF	Developed ex-US Equity	Geographic diversification across Europe, Japan, and Australasia.	**10.0%**	$1,000,000
XLE	Energy Select Sector SPDR	Commodity Equity	Direct stagflation hedge and commodity inflation windfall capture.	**5.0%**	$500,000
XLK	Technology Select Sector SPDR	Sector Growth Equity	Secular productivity growth and technology adoption upside.	**5.0%**	$500,000
EEM	iShares MSCI Emerging Markets	Emerging Markets Equity	Long-horizon demographic growth and global supply-chain beta.	**5.0%**	$500,000

3. Data Sources, Cleaning Discipline & Stationarity Verification

Adhering strictly to Section 9 of the brief, data governance follows institutional standards:
• **Primary Source:** Yahoo Finance (via yfinance) for daily adjusted close prices spanning **2007-01-01 to 2026-10-01 (N=4,966 days)**. Adjusted prices account for dividend reinvestment and stock splits to ensure total return integrity.
• **Risk-Free Rate:** The 13-Week US Treasury Bill yield (^IRX) is annualized and converted to daily continuously compounded rates.
• **Calendar Synchronization:** Trading calendars across international assets are harmonized by forward-filling minor holiday gaps (limit 2 days) then removing non-overlapping dates.
• **Log vs Simple Returns:** Continuous log returns $r_t = ln(P_t / P_{t-1})$ are utilized for econometric stationarity and EVT tail modeling. Discrete simple returns $R_t = (P_t - P_{t-1}) / P_{t-1}$ are used for linear portfolio weighting ($R_{p,t} = \sum w_i R_{i,t}$).

Table 2: Data Cleaning Audit, Outlier Screening & Augmented Dickey-Fuller (ADF) Tests

Ticker	Asset Name	Ann. Mean	Ann. Vol	Min 1D	Max 1D	4σ Outliers	ADF Stat	ADF p-Val	Stationary?
SPY	S&P; 500 ETF	10.38%	19.57%	-11.6%	13.6%	37	-15.87	< 0.0001	**YES (p<0.01)**
EFA	iShares MSCI EAFE	4.71%	21.61%	-11.8%	14.7%	47	-15.44	< 0.0001	**YES (p<0.01)**
EEM	iShares MSCI Emergin	4.85%	27.93%	-17.6%	20.5%	41	-13.25	< 0.0001	**YES (p<0.01)**
AGG	iShares Core US Aggr	2.79%	5.38%	-7.1%	3.8%	29	-14.90	< 0.0001	**YES (p<0.01)**
GLD	SPDR Gold Shares	9.19%	18.23%	-10.8%	10.7%	21	-71.29	< 0.0001	**YES (p<0.01)**
XLE	Energy Select Sector	6.94%	30.26%	-22.5%	15.3%	30	-74.79	< 0.0001	**YES (p<0.01)**
XLK	Technology Select Se	15.60%	23.36%	-14.9%	13.0%	23	-17.58	< 0.0001	**YES (p<0.01)**

Stationarity Conclusion: Augmented Dickey-Fuller tests reject the null hypothesis of a unit root for all 7 assets at the 99% significance level ($p < 0.0001$). No artificial price clipping was performed; extreme tail moves (e.g. March 2020 COVID crash) were preserved as true market tail risk.

4. Layer 1 — Foundations: Portfolio Theory & Institutional KPIs

We construct the Markowitz Mean-Variance Efficient Frontier by solving the constrained quadratic program: $min_w \, w^T \Sigma w$ subject to $w^T \mu = \mu_{target}$ and $\sum w_i = 1, w_i \geq 0$. We compare five candidate portfolio constructions across institutional performance metrics:

Portfolio Name	Ann. Return	Ann. Vol	Sharpe Ratio	Sortino Ratio	Max Drawdown	Calmar	CAPM Beta	Jensen Alpha
Conservative Endowment	**7.89%**	10.40%	**0.611**	**0.870**	**-28.52%**	0.277	0.476	**+1.24%**
Max Sharpe (Tangency)	12.26%	12.35%	0.868	1.251	-27.91%	0.439	0.486	+5.49%
Global Min Variance	3.67%	5.16%	0.413	0.561	-16.64%	0.221	0.070	+1.38%
Standard 60/40 Baseline	8.55%	11.93%	0.588	0.834	-34.70%	0.246	0.600	+0.56%
Equal Weight (1/N)	10.24%	16.28%	0.534	0.756	-42.35%	0.242	0.773	+0.39%

Core Trade-Off Analysis: While the unconstrained Max Sharpe portfolio delivers higher return (+12.26%), it suffers from extreme concentration (>60% in tech/growth) and high volatility (12.35%). The **Conservative Endowment Portfolio** achieves the optimal institutional balance: it produces **+7.89% return** (beating the CPI+4% spending hurdle), generates **+1.24% annual Jensen's Alpha**, and cuts maximum drawdown to **-28.52%** (saving $618,000 in capital preservation vs the standard 60/40 benchmark during crises).

5. Layer 2 — Value-at-Risk (VaR): Multi-Method Comparative Analysis

We implement and compare three distinct Value-at-Risk methodologies across 90%, 95%, and 99% confidence horizons:
1. **Historical Simulation VaR:** Non-parametric empirical quantile of realized portfolio losses.
2. **Parametric (Variance-Covariance) VaR:** $VaR_{\alpha} = -\mu + z_{\alpha}\,\sigma$ assuming Gaussian normality.
3. **Monte Carlo Simulation VaR:** 25,000 synthetic return vectors generated via Cholesky decomposition of the empirical covariance matrix with multivariate Student-t innovations (df=5) to account for joint non-linear tail fatness.

Confidence Level	Parametric VaR (Normal)	Historical VaR (Empirical)	Monte Carlo VaR (t-Copula)	Disagreement Spread (MC vs Param)
90% Confidence	0.808% ($80,800)	0.637% ($63,650)	**0.943% ($94,280)**	+16.7%
95% Confidence	1.046% ($104,600)	0.939% ($93,920)	**1.300% ($129,970)**	+24.3%
99% Confidence	1.492% ($149,230)	1.839% ($183,870)	**2.135% ($213,480)**	**+43.1%**

Critical Econometric Discussion: Where the Models Disagree and Why

1. Failure of the Normality Assumption at High Confidence: At 90% confidence, Parametric VaR (0.808%) exceeds Historical VaR (0.637%) because real asset returns are leptokurtic (peaked near the mean). However, at the regulatory 99% level, the relationship dramatically inverts: Historical VaR (1.839%) and Monte Carlo VaR (2.135%) are **23% to 43% higher** than Parametric VaR (1.492%). The normal distribution severely underestimates tail probability density (excess kurtosis).
2. Window Bias in Historical Simulation: Historical simulation assumes past empirical realizations accurately represent future states. However, historical VaR is a step-function that does not model forward correlation shifts or non-linear co-drawdowns.
3. Monte Carlo Heavy-Tail Incorporation: Monte Carlo simulation with Student-t innovations accounts for simultaneous joint crashes, revealing an institutional 99% daily VaR of $213,480.

6. Layer 3 — Expected Shortfall (CVaR): The Coherent Risk Measure

While Value-at-Risk answers *'How often do losses exceed threshold q?'*, it provides zero information regarding *'How severe are the losses when that threshold is breached?'* Furthermore, as proven by Artzner, Delbaen, Eber, and Heath (1999), **VaR is NOT a coherent risk measure** because it violates the axiom of **subadditivity**: $VaR(X + Y) \leq VaR(X) + VaR(Y)$ can fail in non-elliptical distributions, meaning portfolio diversification can artificially appear to increase measured VaR.

In contrast, **Expected Shortfall (Conditional VaR)** — defined as $ES_\alpha = E[L \mid L \geq VaR_\alpha]$ — is a strictly coherent risk measure. It satisfies subadditivity, monotonicity, positive homogeneity, and translation invariance. Under the Basel III/IV Fundamental Review of the Trading Book (FRTB), regulators mandated replacing 99% VaR with 97.5% Expected Shortfall precisely to penalize catastrophic tail density.

Confidence Level	Value-at-Risk (Historical)	Expected Shortfall (Historical)	Monte Carlo Expected Shortfall	ES / VaR Tail Severity Multiple
90% Confidence	0.637% ($63,650)	1.152% ($115,180)	1.472% ($147,220)	1.81x
95% Confidence	0.939% ($93,920)	1.539% ($153,920)	1.840% ($184,030)	1.64x
99% Confidence	**1.839% ($183,870)**	**2.758% ($275,820)**	**2.784% ($278,400)**	**1.50x**

Institutional Takeaway: When a 99% breach occurs, the expected daily loss is not $183,870 (VaR), but **$275,820 to $278,400 (Expected Shortfall)** — a **50% severity surcharge**. Failing to model CVaR exposes the endowment to severe capital erosion during tail contagion.

7. Layer 4 — Extreme Value Theory (EVT): Peaks-Over-Threshold

To model rare but catastrophic tail risks without relying on ad-hoc distributions, we implement the **Peaks-Over-Threshold (POT)** method. By the Pickands-Balkema-de Haan Theorem, excess losses over a sufficiently high threshold *u* converge to the **Generalized Pareto Distribution (GPD)**:

$$F_u(y) = P(X - u \le y \mid X > u) \approx 1 - (1 + \xi y / \beta)^{-1/\xi}$$

Systematic Threshold Selection (Non-Arbitrary): We scan candidate thresholds across quantiles 90% to 97%. The optimal threshold **u* = 0.0064 (90.0% quantile, N_u=497 exceedances)** was selected where the Mean Residual Life plot exhibits stable linearity and GPD parameter estimates stabilize. MLE fitting yields:

- **Shape Parameter ξ = 0.2440:** Crucially $\xi > 0$, mathematically proving that portfolio losses possess **Fréchet heavy tails** with polynomial tail decay.
- **Scale Parameter β = 0.0039:** Governs the dispersion of tail exceedances.

Confidence Level	Parametric VaR	EVT-POT VaR	Parametric ES	EVT-POT ES	The Tail Risk Gap (%)
90% Confidence	0.808%	0.637%	1.118%	1.154%	+3.2%
95% Confidence	1.046%	0.932%	1.320%	1.545%	+17.0%
99% Confidence	1.492%	**1.845%**	1.714%	**2.752%**	**+60.5% Underestimation**

The Tail Risk Gap: Conventional parametric risk models assume exponential tail decay and severely underestimate 99% Expected Shortfall by **+60.5% ($103,740 per day)**. Extreme Value Theory captures the fat-tail dynamics that actually govern financial crises.

8. Layer 5 — Model Validation: Out-of-Sample Backtesting

To ensure our risk models are not overfitted to historical market conditions, we perform out-of-sample backtesting across the turbulent **2020-01-01 to 2026-10-01 window (N=1,697 trading days)**, which encompasses the March 2020 COVID crash, the 2022 inflation shock, and the 2023-2026 recovery. Models are evaluated using standard regulatory tests:

• **Kupiec Proportion of Failures (POF) Test:** Tests unconditional coverage H_0: $p = 0.01$ using the likelihood ratio statistic: $LR_{POF} = -2 \ln [(1-\alpha)^{N-x} \alpha^{x} / ((1-x/N)^{N-x} (x/N)^{x})] \sim \chi^2(1)$.
• **Christoffersen Independence Test:** Evaluates Markov transitions to detect violation clustering over consecutive days.
• **Basel Traffic Light System:** Regulatory capital adequacy zoning (Green ≤ 4 scaled violations, Yellow 5-9, Red ≥ 10).

Model Architecture	Conf.	Test Obs	Violations	Expected	Violation Rate	Kupiec p-Val	Basel Zone	ES Shortfall Bias
Parametric (Normal) VaR	99%	1695	26	17.0	1.53%	0.0405	**GREEN**	+47.9%
Historical Simulation Va	99%	1695	13	17.0	0.77%	0.3147	**GREEN**	+22.7%
Monte Carlo (Student-t)	99%	1695	10	17.0	0.59%	0.0662	**GREEN**	+34.2%
Extreme Value Theory (EV	99%	1695	**16**	17.0	0.94%	**0.8149**	**GREEN**	+12.8%

Model Validation Findings:
1. **Parametric Model Rejection:** Parametric VaR logged 26 violations (1.53% rate vs 1.00% target), failing with a Kupiec p-value of 0.0405 and suffering from a +47.9% shortfall bias.
2. **EVT Model Superiority:** The Extreme Value Theory (EVT-POT) model registered exactly **16 violations against 17.0 expected** (violation rate 0.94%), achieving a stellar Kupiec p-value of **0.8149** and placing the endowment firmly inside the **Basel GREEN Zone** with zero supervisory capital penalties.

9. Layer 6 — Risk Budgeting & Position Sizing on USD 10 Million

Institutional asset allocation requires looking beyond capital weights to analyze **risk contribution**. We compute Marginal VaR ($\partial VaR / \partial w_i$) and Component VaR ($w_i \times Marginal\ VaR_i$), decomposing total portfolio risk across the USD 10 Million mandate:

Ticker	Asset Name	Capital Weight	Dollar Allocation	Marginal VaR (Daily)	Component VaR ($USD)	% Contribution to Risk
SPY	S&P; 500 ETF	**20.0%**	$2,000,000	2.563%	$51,258	**33.6%**
EFA	iShares MSCI EAFE	**10.0%**	$1,000,000	2.844%	$28,437	**18.7%**
EEM	iShares MSCI Emerging	**5.0%**	$500,000	3.503%	$17,513	**11.5%**
AGG	iShares Core US Aggreg	**40.0%**	$4,000,000	0.207%	$8,288	**5.4%**
GLD	SPDR Gold Shares	**15.0%**	$1,500,000	1.117%	$16,762	**11.0%**
XLE	Energy Select Sector S	**5.0%**	$500,000	3.236%	$16,179	**10.6%**
XLK	Technology Select Sect	**5.0%**	$500,000	2.785%	$13,923	**9.1%**

Strategic Risk Sizing Insights:

• **Fixed Income (AGG):** Represents **40% of capital ($4.0M)**, but generates only **5.4% of portfolio risk**, proving its function as a defensive cash-flow anchor.

• **Core Equities (SPY):** Accounts for **20% of capital ($2.0M)**, but contributes **33.6% of portfolio risk** due to its high correlation with global equity beta.

• **Gold (GLD) & Energy (XLE):** Provide complementary inflation and safe-haven buffers, contributing ~11% and ~10% of risk respectively.

10. Crisis Stress Testing on the USD 10 Million Endowment

To verify resilience during extreme tail dislocations, we subject the USD 10 Million endowment to three historical crisis paths and one custom forward-looking shock:

Crisis Scenario	Historical Window / Narrative	Portfolio Return	Max Peak Drawdown	Dollar Impact ($USD)	Capital Retained
2008 Lehman Shock	2008-09-01 to 2008-11-20 (Interbank collapse)	-21.40%	-22.64%	**-$2,139,553**	$7,860,447
March 2020 COVID Crash	2020-02-19 to 2020-03-23 (Global lockdown crash)	-16.98%	-19.07%	**-$1,698,423**	$8,301,577
2022 Rate-Hiking Drawdown	2022-01-03 to 2022-10-14 (500bps Fed rate hikes)	-15.94%	-16.55%	**-$1,594,042**	$8,405,958
Custom 2026-27 Oil/Rate Freeze	Strait of Hormuz crude +60%, Rates high, Tech -32%, Gold +22%	**-10.75%**	-10.75%	**-$1,075,000**	**$8,925,000**

Stress Test Defense: In the forward-looking 2026-27 Geopolitical Oil Shock, Gold (+22%) and Energy (+38%) contribute +$520,000 in offsetting profits, capping total portfolio drawdown at **-10.75% (-$1,075,000)** and preserving **$8.925 Million** of university endowment capital.

11. Non-Negotiable Assumptions & Practical Limitations

In strict compliance with Section 6, Item 6 of the Official Brief: Plain-Language Disclosure of Model Boundaries

Quantitative models are stylized representations of complex market reality. To maintain intellectual honesty before the Investment Committee, we explicitly state the operational assumptions and blind spots of our risk architecture:

1. Liquidity Freeze & Market Impact Blindness: The models assume all ETF assets can be liquidated at continuous daily closing prices with zero slippage. In a systemic liquidity freeze (e.g. March 2020), bid-ask spreads widen significantly, and selling large blocks of $10M capital would incur price impact costs not reflected in exchange prices.

2. Stationarity and Regime Shift Assumptions: While ADF tests confirm statistical stationarity, financial return generation processes are fundamentally non-stationary. Regime shifts (such as the 2022 transition from ZIRP to 500bps policy rates) break historical covariance matrices, causing correlations between equities and bonds to become positive.

3. Asymptotic Limit Approximations in EVT: Extreme Value Theory relies on asymptotic tail convergence (Pickands-Balkema-de Haan Theorem). While POT GPD fitting outperforms normal distributions, parameter estimation remains subject to finite sample variance and threshold choice sensitivity.

4. Black Swan Unpredictability: Extreme events with zero historical precedence (e.g. sovereign debt restructuring, catastrophic cyberwarfare on exchange infrastructure) cannot be generated by historical simulation or backward-looking copulas.

5. Rebalancing Friction & Custody Drag: Calculations assume frictionless daily compounding without custodial fees, management expenses (5-10 bps ETF expense ratios), or tax drag.

12. Investment Committee Decision Memorandum

Based on extensive econometric estimation, Extreme Value Theory tail diagnostics, out-of-sample backtesting, and crisis stress simulations, the Quantitative Research Division delivers the following formal policy recommendation to the Serendib Board:

FINAL MANDATE DECISION: ACCEPT WITH EVT RISK OVERLAY

1. Fund Allocation Directive: Allocate the USD 10,000,000 capital pool across the 7 prescribed assets: 40% AGG ($4.0M), 20% SPY ($2.0M), 15% GLD ($1.5M), 10% EFA ($1.0M), 5% XLE ($0.5M), 5% XLK ($0.5M), and 5% EEM ($0.5M).

2. Statutory Risk Limits: Enforce a maximum daily 99% EVT Value-at-Risk limit of **1.85% ($185,000)** and an Expected Shortfall ceiling of **2.75% ($275,000)**. Any breach of these thresholds must trigger an immediate committee rebalancing review.

3. Liquidity & Spending Rule Compliance: With an annualized expected return of **+7.89%** and income yield anchored by 40% AGG, the portfolio safely supports the university's 4.0% annual endowment disbursement rule while preserving inflation-adjusted capital in perpetuity.

4. Governance Review Frequency: Perform quarterly GPD threshold diagnostics and rolling out-of-sample Kupiec backtesting to verify model parameter stability.

Signed and submitted on behalf of the Quantitative Research Division for the QuantEdge 1.0 Final Challenge.

13. Academic References & Regulatory Frameworks

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Appendix: Mandatory AI Tools Disclosure (Section 8.7)

In strict adherence to the QuantEdge 1.0 Official Competitor Briefing disclosure requirements:

Item	Required Disclosure Details
AI Tool(s) Used	Google Antigravity / Gemini 3.8 Flash High coding assistant.
Specific Purpose	Code scaffolding, syntax acceleration, Streamlit layout formatting, and ReportLab PDF compilation support.
Human Validation	All mathematical derivations (GPD, MPT, Kupiec POF), statistical calculations, asset weights, and investment recommendations were independently reviewed, audited, and tested by team members.
Material Components	Initial Streamlit dashboard layout was AI-assisted; risk engine calculations, EVT fitting, backtest algorithms, and committee conclusions were independently verified and validated.